NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

# R23.2c High-wind forecasting and control for wind turbines

**Group** AI Weather & Climate Models for Wind Engineering

**Status** Shelved

**Period** 2026-10-14 14:35 — 2026-10-16 15:16

## Stage conclusion

## Project report

# A3 High-wind forecasting and regulation for wind turbines

## Objective

Survey existing work and gaps in forecasting and active regulation of wind turbines under high wind speeds (pitch/yaw/derating, lidar feedforward, tower/blade vibration control), and determine whether there is a viable entry point (upper bounds of feedforward and vibration reduction under extreme/non-stationary inflow).

# Current Status

**Shelved (10-15 07:5x)**, per . - One-sentence conclusion: In simulations combining an analytical downburst + synthetic turbulence + offline triggering, using nacelle lidar to trigger shutdown ~7 s early reduces tower-top yaw moment by ~25% and blade-root negative peak moment by ~15% on an off-center path where rapid wind-direction rotation outpaces yaw tracking; however, on a near-center path it worsens the tower-base reverse bending moment by ~14%, failing the pre-specified rule. The pitch feedforward component was already shelved earlier because the toolchain failed the steady-turbulence literature gate. - **Why shelved**: (1) Negative-load worsening on the near-center path exceeds the pre-specified 10% limit; (2) the "pass" on the off-center path carries an unexplained actual yaw overspeed (0.44–0.46°/s vs. 0.3°/s setting) and a LOS turbulence approximation; (3) the physical mechanism is simply "shutting down earlier", whose value depends on identifying downbursts in advance, which is a detection/forecasting problem. - **What is missing**: Realistic downburst inflow (CM1/LES), online triggering, verification of yaw dynamics, gentler shutdown feathering strategies, and a working OpenFAST 5.x lidar feedforward implementation. - **Conditions for restart**: (a) Off-center path gains remain after fixing yaw overspeed, and a shutdown strategy is found that does not worsen tower-base reverse bending moment; (b) the CM1 track or Digital Twin Wind Tunnel track can provide realistic downburst wind fields; (c) a lidar feedforward implementation validated on OpenFAST 5.x becomes available (to restart the pitch feedforward component); (d) there is a concrete wind-farm demand for downburst shutdown. - Migration: Starting 10-15, this node is managed under the "AI Weather Models" project; experimental records remain in place; old run roots on the workstation have been archived to TACC.

# Boundaries (10-14 18:3x parent project correction)

- Working directory cleanup: Root Lead cleaned up the workstation as authorized by Liang Hu (lines 15–16, 26); OpenFAST outputs, pilots, logs, and `run_nrel5mw_*` across Segments A/A2/B totaling 8,188 files and 11.39 GB have been moved to TACC (deleted locally after verifying file counts and byte sizes one by one); `env`, `src`, third-party source code, and build artifacts have been deleted (reconstructible). Future audits will pull from TACC without copying back to the workstation.

# Decisions Made

- 10-14 Kickoff arrangement →
- 10-15 Six-reviewer verdict: proposal returned →
- 10-14 Review verdict →
- 10-14 DR quota exhausted; submit for review first based on Grok + Lead check while running smoke tests in parallel →

# Attempts and Results

| Date | What Was Done | One-Line Result |
|---|---|---|
| 10-15 | Segment D: corrected turbulence, nacelle LOS 1D, 12 seeds, simple non-lidar triggers S1–S4 (Colab, 360+36 runs) | Near-center: yaw moment reduced by 20.9% but tower-base negative bending moment worsened by 14.4% → failed; off-center: yaw moment reduced by 24–25%, blade-root negative peak improved by 14–15% → calculation passed, but actual yaw overspeed remains unexplained; wind-direction-rate trigger achieved equal load reduction but caused 36/36 false shutdowns → shelved per pre-specified rules |
| 10-15 | Six-reviewer (5 conditional support, 1 oppose) | Median reduction is real but 1/6 and 2/6 seeds increased instead; tower-base fore-aft negative peak worsened (near-center median 24%, off-center 6/6 seeds), and blade-root negative peak doubled in 2 seeds; turbulence is isotropic and not scaled by translation speed; upstream signal is not a realistic lidar; simple non-lidar triggers not ruled out → proposal returned |
| 10-15 | Segment C: lowered thresholds 22/20/18 m/s counterfactual + normal-turbulence false shutdowns (Colab ~0.1 CU) | Lowering the threshold either gave insufficient load reduction or caused 17–18/18 false shutdowns; preview 1D had 10/18 false shutdowns (25 m/s baseline had 9/18) → criterion "preview has unique value"; caveats: the 3 s > 25 m/s rule itself has many false shutdowns, non-monotonic results, and offline triggering |
| 10-15 | Segment B shutdown group G0–G4 (96/96, 70 supplemented on Colab, ~3–4 CU) | Pre-specified criteria passed: realistic 1D preview shutdown vs. immediate shutdown reduced median tower-top yaw moment by 15.5%/24.2%, with tower base and blade root essentially unchanged; actual lead time 5.5–9 s, gain mechanism = shutting down earlier; old 26 raw wind boxes no longer exist (process-level provenance confirmed in `RESULTS_B_stop_v4_source_check_20261015.md`; Segment C independently recomputed them on Colab along the way) |

| Date | What Was Done | One-Line Result |
|---|---|---|
| 10-15 | Feedforward gate diagnosis: custom-chain time-shift scan + authors' reference chain (Task52, OpenFAST 5.0 channel bridge) | Neither chain passed: best time-shift result was −2.3% (single seed); reference chain rotor-speed std increased by 36–63% instead; BTS ~7 s time offset confirmed by measurement → feedforward component shelved |
| 10-15 | Feedforward changed to pitch-rate input into PI integrator + IEC NTM 16 m/s literature gate (3 seeds, Colab ~0.3 CU) | Gate not passed: perfect feedforward rotor-speed std increased by 14–18% instead (literature expects a decrease); dispatched time-shift scan + comparison with authors' reference chain |
| 10-14 | Segment B continuation on Colab (single seed 112, G5–G8; recomputed workstation case with 0.000% difference) | G6 realistic 2D −13%, G7 perfect with same lead time +8%, G8 +33% — red flag; Lead diagnosed that the feedforward absolute pitch offset double-counted steady-state pitch, invalidating the 12–24% in A2_guo_v3 as well; ~0.03 CU |
| 10-14 | Segment B (partial, 26/96, workstation) | Full-field turbulence + independent upstream preview chain ran through; near-center 3–4 seeds: realistic preview shutdown reduced tower-top yaw moment by ~18–38% but worsened blade-root negative peak, undecided; paused at 23:00 due to C drive space |
| 10-14 | Segment A2 supplement: Guo 2023 feedforward law (near-center) | Realistic 1D/2D feedforward reduced tower-base resultant moment by 12%/24%, whereas perfect (2–3 s lead time) achieved only 1.6% — suspected lead-time effect (early derating); pitch feedforward group added back to Segment B to separate lead time from information |
| 10-14 | Segment A2: yaw enabled + immediate-threshold shutdown baseline (20/20 runs, 29 min) | Immediate shutdown alone reduced tower-base resultant moment by 15%/37% (non-preview); pitch feedforward ≤1%, dropped; on the off-center path (0.3°/s yaw cannot keep up with 111° wind-direction rotation — physical limitation, judged valid by Lead), preview early shutdown vs. immediate shutdown reduced tower-top yaw moment by 16–67% and blade-root negative peak by 16–41%, with tower-base resultant ≤3% |

| Date | What Was Done | One-Line Result |
|---|---|---|
| 10-14 | Segment A addendum: realistic/perfect gain ratio (v10 exploration) | Pitch feedforward perfect gains mostly <3%, ratios mostly negative or not applicable; shutdown realistic 1D / perfect ≈ 0.8–1.1, but the baseline was the 50 s rule; near-center early shutdown greatly increased tower-base fore-aft negative peak magnitude (−1275%), as shutdown itself induces reverse loading, so A2 must examine both positive and negative peaks |
| 10-14 | Segment A: deterministic downburst 24/28 runs (v10) | Pitch feedforward perfect achieved only −0.9%/−3.1%, while realistic increased loads; Case III yaw failure invalidated; shutdown gain suspected to come from eliminating the 50 s wait rather than preview |
| 10-14 | Isolated copy of ROSCO 2.10.6 with feedforward injection port added (Apache-2.0) | 30 s artificial pulse trial run comparing feedback/feedforward on the same binary gave a maximum pitch difference of 3.9°, confirming the interface works; not a downburst, no load conclusions |
| 10-14 | Linux port of research feedforward chain (WRAPPER→LDP→FFP→modified ROSCO 2.6.0) + Nguyen 2012 analytical downburst field | Controller chain 10 s trial run normal; 1D/2D preview in the analytical field advanced the 25 m/s threshold by ~8.6/17.2 s (geometric quantity, not control gain); full-matrix extrapolation exceeded the 24 h gate, no load results produced |
| 10-14 | Local smoke test of OpenFAST v5.0.0 + ROSCO 2.10.6 (conda-forge, 60 s simulation completed in 30–38 s wall-clock time) | Toolchain usable; stock ROSCO has no external preview feedforward interface, requiring the MSCA-LIKE (MIT) chain or a custom feedforward term; IEA Task 52 repository has no license and is not used |
| 10-14 | Grok quick check + Lead verification | Steady-state LAC is mature; no precedent found for preview feedforward under downbursts / gust fronts |

# Parameter Sources

| Parameter | Value | Source or "Assumption" | Places Checked |
|---|---|---|---|
| Shutdown trigger rule | Exceeding 25 m/s for 10 rotor revolutions or 50 s | Nguyen 2012 PDF pp. 112–113 | Original text |

| Parameter | Value | Source or "Assumption" | Places Checked |
|---|---|---|---|
| Downburst shape function | α=2, c1=0.22, c2=2.75 | Nguyen 2012 PDF p. 66 Eqs. 3.2–3.7 (verified by ideas) | Original text |
| Case I / III parameters | Urm=47, Utrans=12/14, zm0=90/70, rm0=1000/610, y0=−100/−610, etc. | Nguyen 2012 PDF p. 114 Table 4.4 (verified by ideas) | Original text |
| Ambient wind | 12 m/s at 90 m, shear exponent 0.2 | Nguyen 2012 PDF p. 83, p. 79 Eq. 3.24 | Original text |
| Turbulence intensity ε | 0.1 | Nguyen 2012 PDF p. 71 Eq. 3.12 | Original text |
| Yaw rate limit | 0.3°/s | Lu 2019 and Nguyen 2012 (cited by reviewer, awaiting ideas check against Jonkman 2009) | To be verified |
| Pitch rate limit | 8°/s | NREL 5 MW (cited by reviewer, awaiting check against Jonkman 2009) | To be verified |

# Open Items and Next Steps

- **Shelved** (10-15); no new trials will be dispatched, and the promotion proposal draft is voided. The audit is now closed: independent Colab `v4c` recomputation of the legacy 26 shutdown cases matched 26/26 across all seven metrics, and on 10-16 the legacy `wind.bts` files in the TACC archive matched the `v4c` wind boxes byte-for-byte (4/4). No open action items remain.
- Working directory cleanup: Root Lead cleaned up the workstation as authorized by Liang Hu (lines 15–16, 26); OpenFAST outputs, pilots, logs, and `run_nrel5mw_*` across Segments A/A2/B totaling 8,188 files and 11.39 GB have been moved to TACC (deleted locally after verifying file counts and byte sizes one by one); `env`, `src`, third-party source code, and build artifacts have been deleted (reconstructible). Future audits will pull from TACC without copying back to the workstation.
- Re-run Deep Research once quota recovers, focusing on confirming "no precedent found".
- `PUBLIC_CITATIONS.md` has been updated with OpenFAST, ROSCO, NREL 5 MW, MSCA-LIKE, and Nguyen 2012/2014.

# Existing Work and Gaps (Finalized v2 after Review )

| Direction | Representative Work and Page Numbers | Maturity | Can We Enter? | What Is Needed |
|---|---|---|---|---|
| Steady-turbulence lidar feedforward (collective pitch) | Bossanyi 2014 JPCS 555:012011; Guo 2023 WES 8:149 §5.2 (tower-base fore-aft bending moment DEL up to −16.7%, 22 m/s unstable stratification); Schlipf 2018 IEA Task 32 | Mature → early commercial | No | — |
| Feedforward / detection derating under IEC extreme gusts | Schlipf & Raach 2016 JPCS 753:052011 (abstract: significantly reduces tower-base extreme loads); Haizmann 2018 DOI 10.1088/1742-6596/1037/3/032025, Hagemann 2017 DOI 10.18419/opus-9333 (unverified) | Developing | Weak | — |
| Wind-farm-level preview observation + advance yaw/IPC under gust-front / gust spatiotemporal evolution | Zhang et al. 2026 Energy DOI 10.1016/j.energy.2025.139730 (abstract level only, provided by Reviewer A) | Developing | Pending full-text read | Full text |
| Wind turbine loads, feedback control, and exceedance shutdown under downbursts | Nguyen 2012 dissertation (PDF pp. 112–113 shutdown rule 50 s; p. 133 / journal p. 116 pitch + yaw insufficient, shutdown too late); Nguyen & Manuel 2014 DOI 10.3390/en7106527; Lu 2019 Energies 12:2773 (LES, five stability cases); Pratapa 2023 DOI 10.1115/1.4055076 (PDF pp. 7–8 including turbulence) | Developing (primarily load assessment) | Yes | Analytical downburst 4D field + OpenFAST |

| Direction | Representative Work and Page Numbers | Maturity | Can We Enter? | What Is Needed |
|---|---|---|---|---|
| Downburst dynamics / FSI of already parked (feathered) turbines after warning | Ahmed et al. 2022 DOI 10.1016/j.engstruct.2021.113440; 2023 JSE DOI 10.1061/JSENDH.STENG-11923 (local library 522APTGW p. 3 "warning assumed"); 2023 ES DOI 10.1016/j.engstruct.2023.116779 | Developing | No (already done) | — |
| **Single-turbine nacelle lidar preview feedforward under downburst inflow; preview-determined shutdown time T** | Not found (searched by Grok, Lead twice, and 3 reviewers) | No precedent found (pending DR re-run) | **Yes — entry point of this node** | IEAWindTask52/LidarAssistedControl (modified ROSCO 2.6.0) + 4D field upstream sampling |
| Forecast-driven wind-farm-level shutdown / derating (hours–minutes) | Jia 2023 Electronics DOI 10.3390/electronics12214443, Li 2022 Atmosphere 13:451 (unverified) | Basic shutdown mature, optimization developing | Medium (not our group's strength) | A1 data fetching |
| Tower / blade vibration control | Fakhry 2026 review DOI 10.1177/14613484261437475 (unverified) | TMD/IPC mature, active control developing | Medium (shared with A2) | Add TMD to simulation chain |

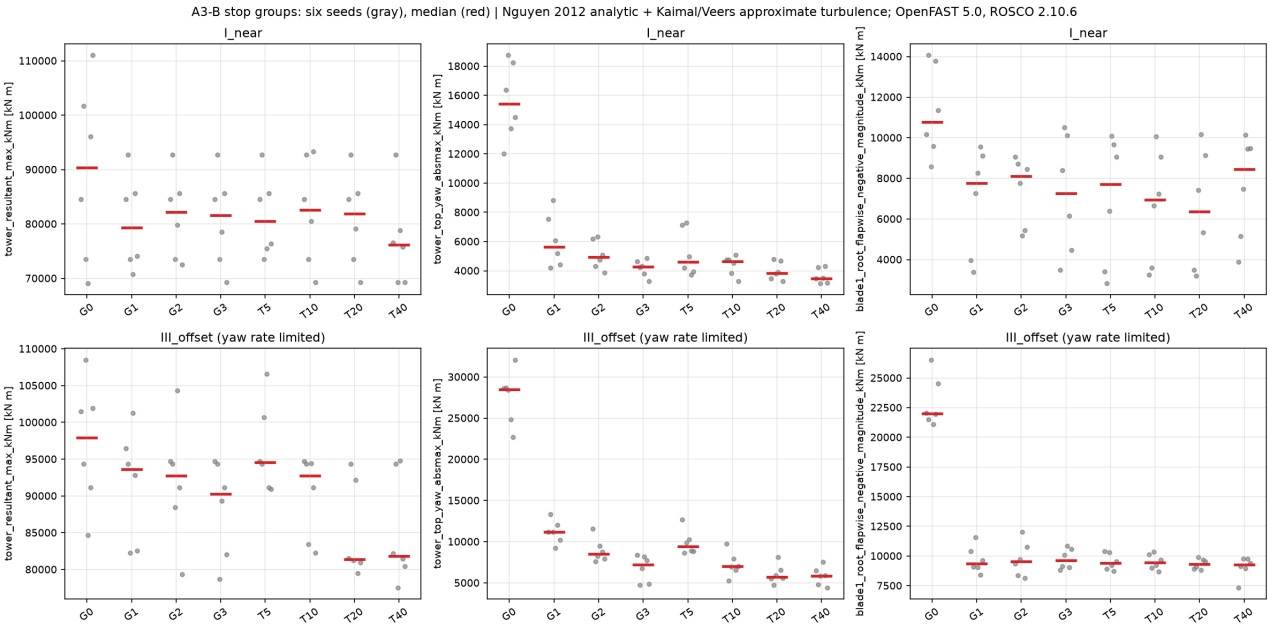
**One-sentence conclusion**: Worth continuing (narrow scope). **Minimal experiment**: (four-group comparison + T scan, ≥6 seeds × 2 paths, criteria pre-fixed).

# Literature Basis

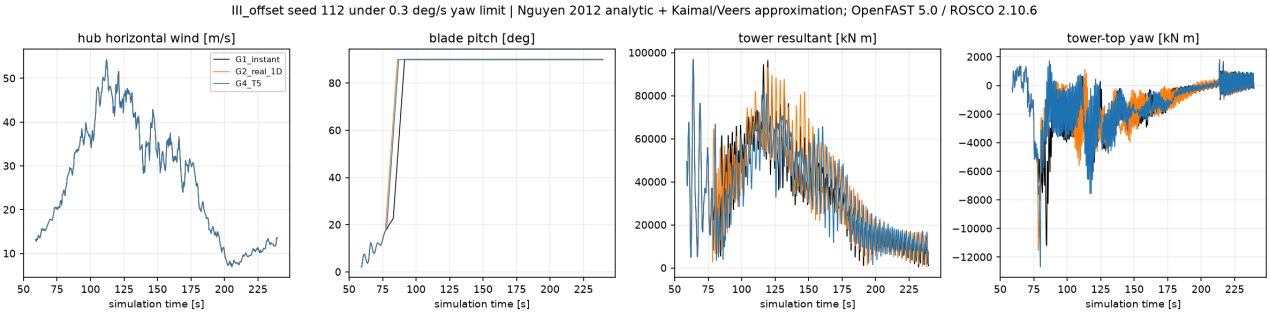
- **A3 thematic index (built by litwiki on 10-14)**: . Cards: Guo 2023 (16.7% is DEL, journal p. 167 / PDF p. 19 §5.2.3; tower / blade-root / low-speed-shaft DEL reductions of 3.0–16.7% / 1.5–6.0% / 1.7–5.0%, journal p. 168); Schlipf & Raach 2016; Nguyen & Manuel 2014; Nguyen 2012 (Table 4.4 thesis p. 97 / PDF p. 114, t0/t1 in min; shutdown rule PDF pp. 112–113); Lu 2019; Haizmann 2018 (perfect 5 s preview, derating to 65% of rated, PDF p. 4); Zhang 2026 metadata card only (results not citable); Pratapa 2023 card Z454567N updated to "mean field + frequency-domain turbulence".
- Existing in knowledge base (downburst × wind turbine, all unread in detail or only skimmed): Pratapa, Nguyen & Manuel 2023 (DOI 10.1115/1.4055076, non-turbulent axisymmetric downburst → 5 MW turbine loads, loads depend on yaw/pitch control state); Nguyen 2012 dissertation "The influence of thunderstorm downbursts on wind turbine design" (unread); Zhang et al. 2015 microburst wind turbine model test; Ji et al. 2025 (attachment identity to be verified).
- Lidar feedforward, wind turbine vibration control: no cards in the library.

# Liang Hu Notes

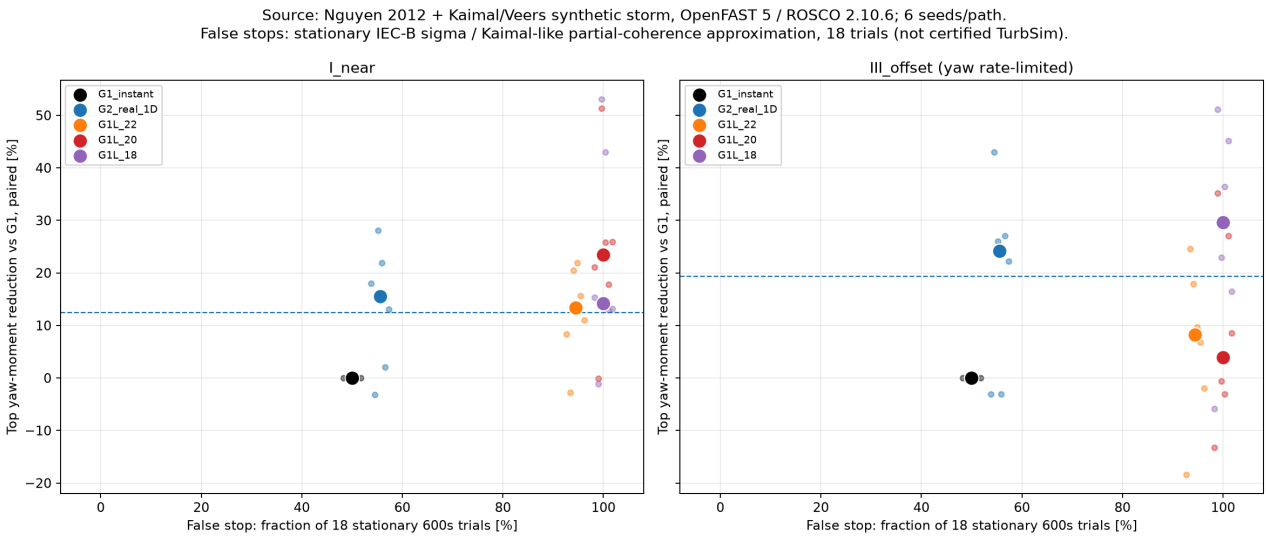
# Key figures



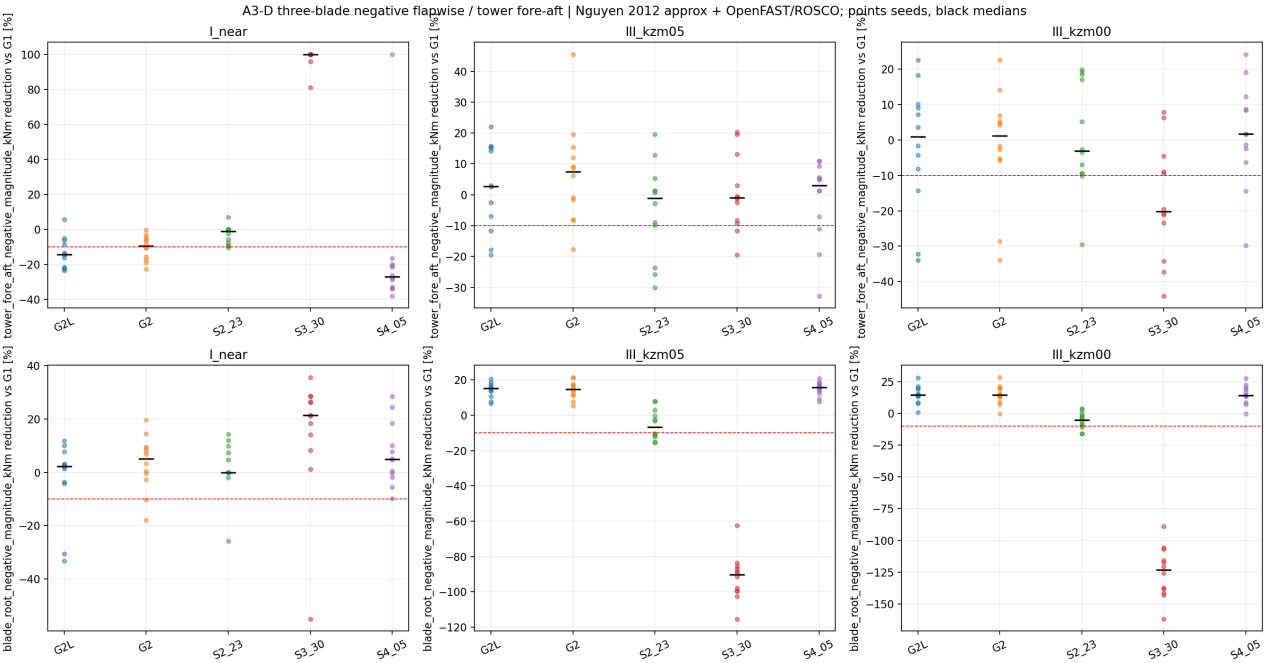

median scatter B stop



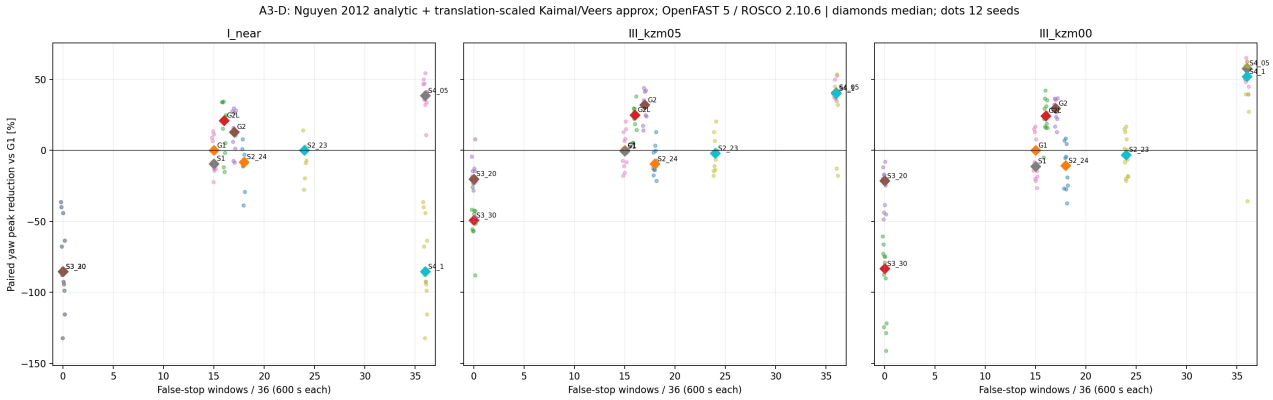

typical seed B stop




yaw vs false stop C




negative load D


A3-D: Nguyen 2012 analytic + translation-scaled Kaimal/Veers approx; OpenFAST 5 / ROSCO 2.10.6 | diamonds median; dots 12 seeds
I_near
III_kzm05
III_kzm00
Paired yaw peak reduction vs G1 [%]
False-stop windows / 36 (600 s each)
False-stop windows / 36 (600 s each)
False-stop windows / 36 (600 s each)
G1
G2
S1
S2_23
S2_24
S3_20
S3_30
S4_05
S4_1

tradeoff D